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Multilingual Sentiment Analysis of YouTube Live Stream using Machine Translation and Transformer in NLP

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Abstract— **YouTube has become one of the all-inclusive video streaming sources on the internet. Today, the news is streamed on YouTube, marketing of a product is done live on YouTube and it has become a platform for one of the biggest PR producers for companies. Various companies have proposed an optimized way of understanding and getting the opinions of the viewers from YouTube live chat and find the best possible way to provide relevant and informative content to boost the business strategy. This study uses Natural Language Processing (NLP) based approach along with NLP transformers to classify and analyses the sentiment.**

Keywords— YouTube, live stream, live chat, transformer, multilingual, sentiment analysis

I. Introduction

Social media apps such as Facebook, YouTube, Instagram, Google+, and many more are used by millions to share and receive information in different mediums (text, photos, videos, and so on.) [1]. The increase in data exchange calls for real-time analysis. According to YouTube, India already has 245 million active users, reaching 80% to 85% of the cyberspace utilizing population [2]. The domain of video telephony is expected to hit $330.51 billion by 2030. (Grand View Research). YouTube is a major platform used for video sharing, among many other prevalent platforms. It hosts a multimodal, multilingual, multi-domain, and multi-cultural environment with various functionalities. Due to this versatile nature, the utilization of YouTube by the corporate and academia communities is steadily growing.

In order to increase user interaction, YouTube's live stream has live chats, allowing the users to interact with the world by expressing their opinions through chats and comments [1]. Moreover, opinion-mining uses natural-language processing categories such as positive, negative, and natural. Sentiment analysis can be defined as the process of classifying the user's opinion on different topics. The emotions on the specific topic e.g., a viewer is happy watching the video or the content in the video, and their reaction to the video refers to the fact that user is satisfied with the content [3]. Similarly, the same concept can be applied while the viewer is criticizing any content. These analysis helps to understand the viewer reactions to various kinds of content and help the companies to improve their service according to the viewers' demand.

The study uses NLP based approach to understand sentiment analysis on a live chat of a YouTube live stream to retrieve the most relevant and ideal live stream content to get the appropriate result. This process works in a few steps, such as extracting and pre-processing to prepare for the following process, followed by translating the language from non-English to English and generating the dataset [1] [2]. Finally, the NLP transformer is used to classify sentiment and get the insight.

For the purpose of this study, metadata is collected from YouTube live streams. According to the URL link, the data is extracted with the help of a data scraping library. However, the extracted chart is heterogeneous in terms of language and various viewers' notions. Therefore, the study performs some pre-processing with the unstructured data to generate the required data set and handles non-English languages with machine translation [4]. Text analysis uses advanced machine learning methods like Deep Learning and Affective Computing. With the help of a pre-trained transformer model, the study analyses the sentiment of the YouTube live stream. Sentiment Analysis using NLP Transformers BERT – developed in 2018. However, in this paper, BERT analyses the sentiment of the multilingual YouTube live chats in a real-time scenario with the accuracy of 85%.

The rest of the paper is organized as follows. The literature review is narrated in the second section. In the third section, the proposed framework is explained. The methodology is explained in the fourth section, the experimental setup and the conclusion is given in the last section.

II. Literature Review

Hanif Bhuiyan et al., [1] have identified a problem, that the ratings of a video is influenced by the relevance of the video, its quality, and popularity. Some irrelevant and low-quality videos are ranked higher which is an indirect result of the number of the likes and dislikes and the number of views it receives. It used an NLP based approach for sentiment analysis to address this problem. The research used web API through HTTP GET function. They created two datasets in which they tested and trained and achieved 75.235% accuracy. Tao Li et al., [4] presented, the development method, content, and data of the YouTube AV 50K dataset which is a freely accessible compilation of more than 50,000 YouTube comments and associated metadata, highlighting its potential applications. To analyse the dataset, comprehend public perceptions toward self-driving cars, and understand public reactions to the tragedy, they conducted a case study of the first fatal accident involving a self-driving car. They outlined and assessed the dataset's performance as well as some of its potential uses.

Ganpat Singh Chauhan and Yogesh Kumar Meena [5] studied the Aspect-based sentiment analysis which was used to examine the influence of video relevant elements on user comments during video retrieval. Their approach was to apply ABSA on natural language processing to user comments in order to increase video retrieval rankings and tokenize each sentence. It was noted that after pre-processing, the topics from the English comments were enriched using the SentiWordNet module which improved the relevancy and quality of the video content. Ekki Rinaldi and Aina Musdholifah [6] proposed the FVEC method for pre-processing and identifying the optimum kernel function for accurate opinion mining on YouTube comments among Indonesian users. This study examined the linear, polygyrid 2, polygyrid 3, and RBF kernel functions. According to the experimental findings, when compared to degree 2 polynomial, degree 3 polynomial, and RBF kernel functions, FVEC-SVM with a linear kernel function has the best performance.

Shanta Ramaswamy et al., [7] proposed a way to obtain and examine the information rendered in a video, to decide what should be the contents of the video. To detect the polarity of the videos, an approach that is based on sentiment analysis was used. A limited data collection was used to test the system. To automate the procedure, machine learning ideas including neural networks, genetic algorithms, SVMs, and Bayesian learning were used. Sonali Rajesh Shah et al., [2], first subjected the dataset to sentiment analysis (SA) in Marathi (Marathi + English) and retrieved Devanagari Marathi comments from YouTube API of major Marathi channels. Following this, various machine learning models and three different vectorization approaches were used in the dataset. The Marglish dataset surpassed the MLP and Count Vectorizer machine learning algorithms, although BNB fared impressively, with Count Vectorizer performing even better. The datasets were subjected to deep learning and more sophisticated deep learning algorithms. The labelling process facilitated the use of clustering techniques such as density-based spatial clustering of applications with noise (DBSCAN) and K-means. Emoticons were considered while extracting a new dataset. The data was then examined for various genres, such as music, education, etc.

Abdullah O. Abdullah et al., [8] conducted a benchmark study of standard filtering techniques for YouTube comment spam. The study used datasets collected from YouTube using API. The study compared many methods including genetic, machine learning, and statistical algorithms. Testing was performed at WEKA and a few additional libraries using ready-made algorithms. The rate of TP proved to be a setback with the exception of AGA, ICA and k-NN. Rawan Fahad Alhujailiet and Wael M.S.Yafooz [9] investigated the sentiment analysis approaches and procedures that were applied to YouTube videos. There were some pre-processing steps that included normalization, stemming, tokenization, and stop word removal. They used ML algorithms or lexicon methods and aspect-based sentiment analysis (ABSA) approach for sentiment classification and extracted in 3 types of SA, namely, simple SA, complex SA, and advanced SA. For sentiment analysis, pre-processing steps were performed in WEKA using the "StringToWordVector filter" and the TF-IDF model to convert each tweet into a word vector form. For stemming, the Snowball Stemmer library was used.

Lukas Stappen et al., [10] explored a lexical technology enumerating an avenue to obtain interpretation from video transcriptions of a considerable multimodal dataset (MuSe-CAR). They used SenticNet to abstract natural language concepts and fine-tune various feature types on a subdivision of MuSe- CAR. With these features, the information of a video and study to forecast helped analyze emotional valence, arousal, and announcer topic classes. They used the extracted features to instinctively categorise video chunks in terms of arousal and valence, as well as ten area specific speaker topics. Long Mai and Bac Le [11] proposed a framework for instinctively acquiring, filtering, and analysing YouTube comments for a given product. They have used joint approach and aspect level sentiment analysis (ASA) in which many aspects of sentiment analysis were derived into opinion target extraction (OTE) and sentiment polarity classification (SPC). In the case of time series data analysis, they have used gated recurrent unit. Bidirectional Encoder Representations from Transformers (BERT) was applied to other problems in NLP.

III. PROPOSED FRAMEWORK

The proposed architecture as shown in Fig.1 consists of two phases; translating non-English text to English and classifying the sentiment of the data with the help of transformer. The pre-processed data was taken for the language translation where it translated the non-English text to English using Googletrans package which utilized Google Neural Machine Translation (GNMT). Furthermore, the second phase classified the sentiment of the data with the help of transformer where it has put the labelled data into the dedicated NLP pipeline to train the transformer model. By the completion of training, the model was tested and validated with live chat pre-processed data, and used evaluation matrices to measure the accuracy and evaluate the model's performance.

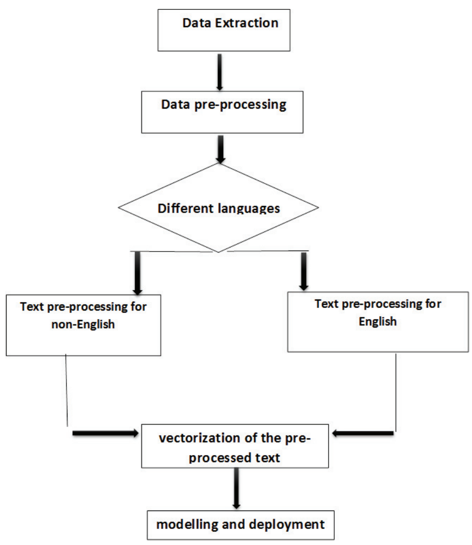

Fig. 1. Architecture for Multilingual Analysis

IV. METHODOLOGY

This segment puts forth the NLP-based approach of sentiment analysis on YouTube live chat of most appropriate and ideal YouTube videos [1]. The preferred frameworks in five stages are as demonstrated in Fig.1. The first step would be aggregation and pre-processing of data with the help of various libraries, followed by transforming multilingual units using Googletrans. The second step is to apply the Bert model to tokenize the dataset and vectorize it into a good numerical measure to characterize the textual data. And the last step in the process is to apply the cased version of Bert model classifier standard deviation (SD) to achieve the accuracy and f1score. The methodology is interpreted in detail below.

A. Collection of data

This step aims to obtain the live chat of the YouTube live stream to fit the data into proposed pipeline. In order to collect the Metadata in the panda's data frame, a library namely chat-downloader was used. In accordance to the live stream URL, it extracted live chat of the videos through Chat downloader. After extracting the chat, following changes were performed:

A dedicated chat scrapper library in python called chat-downloader was used to scrape the data from the live stream video chat of YouTube, followed by input of the URL of the YouTube live stream. Then it collected all the chats with the user name, time, date, and comments. It was then converted into a data frame using pandas' library with the help of looping statements, which ended up with the data set with only one column. Furthermore, with the help of regular expressions, it derived the user's name and deleted the time and other data from the data; after which it provided the data for the columns which has the chat and names of the viewers. At this stage, the data had to be pre-processed.

B. Multilingual to English language

There are many languages other than English that are used in the live chats, where one can use a translator that helps us to convert any source language to a target language. This study aimed to convert any language to English, for which 'Googletrans' was used, which is a free python library and it helped to implement Google Translate API. The content column was then converted in the form of a list and iterated in a loop where each row itself is a list element from where it is captured by the translator and gets converted into a English language.

C. Data pre-processing

Many libraries are used in this process, the main one being the NLTK (Natural Language Tool Kit) which helps to work efficiently on the text data and lower casing the entire content column. The URLs from the data is then removed along with the punctuations and stop words. Similarly, the chat conversion (for an example lol to laugh out loud) and emoticons to text were also performed.

D. Sentiment Analysis using NLP Transformers BERT

BERT (Bidirectional Encoder Representations from transformers) understands the text by going through the previous words and the next words. Particular attention is required to present the transformer model. Hence the model will take the entire sequence of the tokens all at once [12].The attention modelling helps to understand the circumstantial relation between the words, where it has a way for encrypting on the basis of the word context or meaning, wherein a single word will have multiple meaning. This will also help in masking and in producing the next sentence. BERT is clearly a pre-trained stack of transformer encoders. Machine learning models do not work with the raw text that are essential to convert the text into numbers [13]. In order to achieve that, tokenizing the sparest sentence and sentiment classification must be carried out. Passing the text of consent length (padding) and generating an array of 0s (pad token) and 1s (real token) which is attention masking through cased and uncased model. Cased model was opted to get a better understanding of the emotions in a text. (WHAT and what are different emotions). Cased model understands the sensitivity of the text. For the tokenization, this study opted pre-trained tokenizer, BertTokenizer. There are many special Tokens which help the BERT model in classifying, padding, and understanding the unknown tokens.

These all were performed using encode plus. The tokens were then encapsulated in a tensor and padded to a length. Our model worked only with fixed-length string that stored the token length (Fig.2,3,4,5) of each chat message.

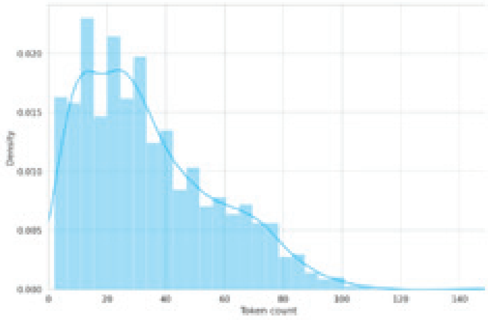

Fig. 2. Pre-Defined Dataset

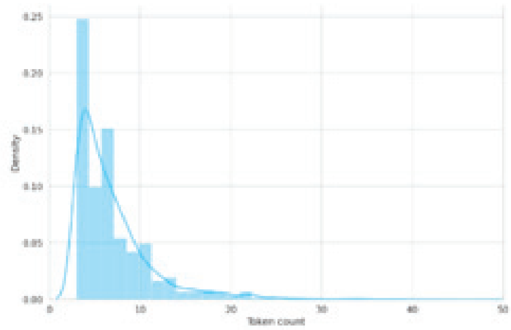

Fig. 3. Length of the Tokens in the Live Chat- Data Set -1

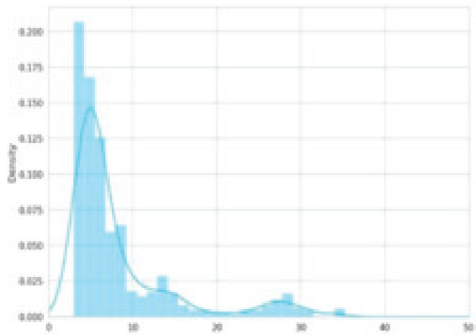

Fig. 4. Length of the Tokens in the Live Chat - Data Set -2

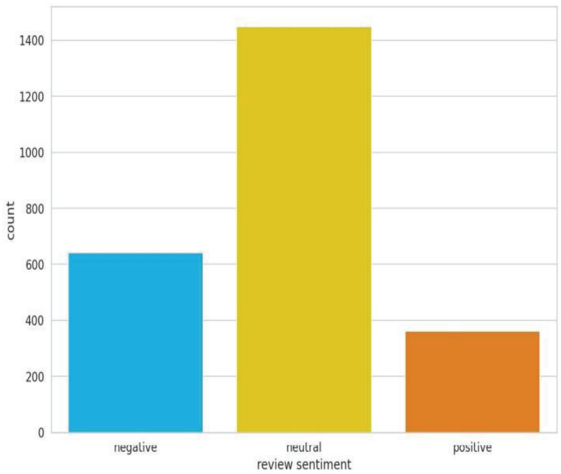

Fig5. Bar-Graph of Sentiments

The length of the tokens can vary from video to video and as a safety measure, the max length was fixed at 160. A chunk was then built to generate the PyTorch dataset and the tokenizers performed the heavy lifting. Finally, the dataset was split into train, test, and validation dataset. Where test and train split ratio 0.1 and the split ratio of validation set is 0.5.

Therefore, our model was trained with a predefined dataset text messages and had the target as sentiments i.e., (positive, negative, natural) and had Label Encoded the target variables. It generated a couple of data loaders, such as train data loader, validation data loader and test data loader with the batch size of 16. where input_ids and attention_mask is of size [16,160] and targets shape is 16.

Basic BertModel was picked and built a sentiment classifier on top of it, where there were 768 hidden unit in feedforward networks [14]. The experiment then created a sentiment classifier that used the Bert model. The classifier represented most of the lifting to the model that had a dropout layer for the regularization and a fully connected layer for the output. At the last layer raw output is returned for the cross-entropy loss function in PyTorch. It generated an object and shifted it into the GPU. For the prediction probabilities of our trained model SoftMax function was used. The AdamW optimizer which was provided by Hudding face was used to correct the weight decay and linear scheduler with no warmup and fine-tuned batch size, learning rate, number of epochs.

Whenever a batch is fed into the model, the scheduler is called. Another helper function to evaluate model to provide the data into the loader. A training loop was used to store the training history and provided the epoch size of 10. Also, an optimizer is used called AdamW which has all the model parameters with learning rate of 2e-5, correct_bias (it helps whether to correct bias in Adam) is put to false and there were many other parameters which were taken as default parameter values. After training our model with predefined data set, our model used the data from live chat messages and labelled their sentiments that were then converted the as per the data frame and fed the output data to our model to the evaluate performances. The epoch size used for the experiment was 10. It then calculated the test accuracy on our test data using a prediction function and an evaluation matrix.

V. EXPERIMENTAL SETUP AND RESULTS

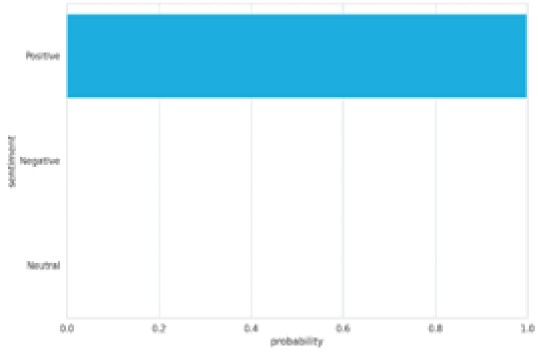

Fig.6. Probability of Sentiment Classes

The probability of the sentiment of the sentence in Fig.6 which was passed as the validation sample sentence says that the probability of the given sentence is positive for 1.0 per chance which indicates that it is positive sentence.

For the evaluation of our model, the study ran a confusion matrix (see Fig.7), where it was observed that the model performed as expected in classifying the negative and positive whereas it did not do well in classifying the natural.

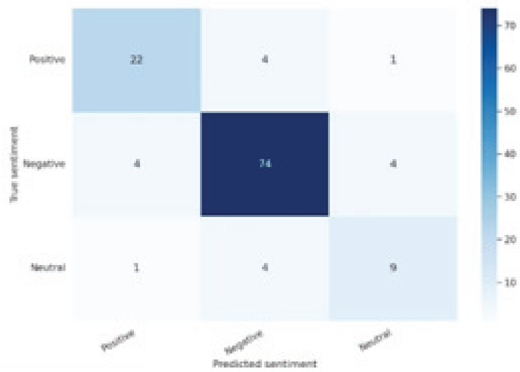

Fig.7. Confusion Matrix

Table 1 is the comparative study of the leading alternative methods for F1 Score and Accuracy.

TABLE I. COMPARATIVE ANALYSIS

Model	*F1 score*	*Accuracy*
FVEC-SVM	68%	62.76%
ABSA	71%	74%
LIWCO7	58.81%	66.47%
StanfirdDM	60.45%	60.46%
Bert Transformer	64%	85%

It is clearly observed that the combination of dedicated NLP pipeline and Bert cased model with sentiment classifying layer also AdamW optimizer. which have resulted with 85% of test accuracy.

Evaluation matrix explaining the type one and type two error in the form of precision and recall

i.e.

Precision = True Positive / True Positive * False Positive

Recall = True Positive / True Positive * False Negative

F1 score = 2* Precision* Recall/ Precision + Recall

In the Table 2, the outcomes of the evaluation matrix is presented. The model has been classified with respect to positive, negative and neutral sentiments. F1 score balances out the perception and identifies the best measure of accuracy.

TABLE II. EVALUATION MATRIX

	Precision	*Recall*	*F1-score*
Positive	0.81	0.81	0.81
Negative	0.90	0.90	0.90
Neutral	0.64	0.64	0.64
Accuracy			**0.85**

VI. CONCLUSION

This paper exemplifies the extraction of pure emotion in the form of sentiments and analysis frequency of one's opinions. Sentiment analysis is carried out on YouTube live stream chat with the help of NLP transformers called BERT model. The study takes into account a user defined dataset to train the model and proceeded with the scraped live chat and retrieved accurate results. F1 score has increased with the proposed method and helped retrieve an efficient classification of the sentiments of multilingual chat.

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